Explicit-solvent electrostatic dephasing in tandem Venus

Standalone corrected-coupling sensitivity note

Tandem-Venus project

21 July 2026

Abstract

This note records a high-cadence explicit-solvent QM/MM electrostatic-gap test performed as a sensitivity analysis for the tandem-Venus exciton model. Two contiguous 4 ps NVT segments were sampled every 4 fs, and a STEOM-CCSD excited-minus-ground difference-density probe was used to measure environmental gap fluctuations at both chromophores. A classical second-cumulant treatment gives an inter-site pure-dephasing time of 21.8 fs; an independent OpenMM PME finite-difference calculation gives 24.0 fs. The local protein dominates the differential noise, whereas water is slower and more common-mode. These values are useful as a sensitivity bound, but they are not a rigorous quantum-bath prediction of an experimental coherence time. The results are deliberately kept separate from the current manuscript.

Status. This is a technical calculation note, not a manuscript revision. In particular, the manuscript's phenomenological $T_2^* = 60$ fs reference has not been replaced by the classical estimate reported here. Earlier versions of the coupling-dependent spectral example predated the 7 August 2026 reciprocal-length unit correction. The spectral section and figure in this version have been regenerated from the corrected production value $J = 32.82 \pm 1.55\ \mathrm{cm}^{-1}$ documented in `reference/orca_validation.json`; the independently calculated electrostatic-gap and dephasing-time results are unaffected.

1 Question and simulation record

The calculation asks whether equilibrium fluctuations of the explicit protein, water and ions can generate dephasing on the same timescale as the solvation memory used in the physical discussion of tandem Venus. The retained full-solvent model contains 69,471 atoms. Starting from the 1 ns production checkpoint, two contiguous 4 ps NVT segments were propagated with a 2 fs timestep. All atoms were saved every 4 fs, giving 2,000 frames over 7.996 ps. Both chromophores used the retained AMBER bonded terms and exact anionic CR2 charges.

This trajectory resolves the observed ultrafast decay and permits independent segment and 1 ps block tests. It is not long enough to converge slow protein or bulk-solvent modes. Consequently, the analysis below is an ultrafast equilibrium-fluctuation test rather than a complete solvation-response calculation.

2 STEOM difference-density probe

The S1 excited-minus-ground density was reconstructed from all seven printed STEOM natural-transition-orbital (NTO) pairs. Their summed represented occupation is 0.99045993. Each particle and hole orbital was normalized on its cube grid before forming

$$\Delta\rho(\mathbf{r}) = \sum_{k=1}^{7} n_k \left(|p_k(\mathbf{r})|^2 - |h_k(\mathbf{r})|^2\right). \tag{1}$$

The resulting neutral density was partitioned onto the 41 *physical* QM atoms. Hydrogen caps and other link atoms were excluded from the partition and therefore carry no difference-density charge. This exclusion is also applied to density visualizations: the plotted probe contains only physical atoms.

For a chromophore site $m \in \{A, B\}$, the environmental contribution to the vertical gap was evaluated as

$$\delta E_m(t) = \sum_{i \in \text{probe}} q_i^\Delta \Phi_{\text{MM}}(\mathbf{r}_{mi}, t), \tag{2}$$

where q_i^Δ is the atom-partitioned difference charge and Φ_{MM} is the instantaneous electrostatic potential from the explicit environment. The probed site's own CR2 and its stacked Tyr residue were excluded from the environmental sum to avoid self-interaction. Separate protein, water and ion traces were retained.

3 Correlation and second-cumulant coherence

The relative phase of the two local excitations is driven by the differential gap

$$X(t) = \delta E_A(t) - \delta E_B(t), \tag{3}$$

not by either absolute site gap alone. After subtracting the mean, its stationary correlation function is

$$C_X(t) = \langle \delta X(t)\, \delta X(0) \rangle\,. \tag{4}$$

In the classical Gaussian second-cumulant approximation, the inter-site coherence envelope is

$$g(t) = \frac{1}{\hbar^2} \int_0^t (t - \tau) C_X(\tau)\, d\tau, \tag{5}$$

$$\frac{|\rho_{AB}(t)|}{|\rho_{AB}(0)|} = \exp[-g(t)]. \tag{6}$$

The reported T_2^* is the first time at which this envelope reaches e^{-1}. Solvation memory and coherence decay are therefore related but are not the same quantity: the former describes decay of $C_X(t)$, while the latter accumulates both its amplitude and memory through the time integral in Eq. (6).

4 Numerical results

Table 1: Eight-picosecond electrostatic-gap results. Correlation and lifetime entries are first $1/e$ crossings.

Environment	σ_{A-B} (cm^{-1})	$\rho_{AB}(0)$	Bath memory (fs)	T_2^* (fs)
All MM charges	385.0	0.065	61.3	**21.80**
Protein + water	387.2	0.069	62.1	21.62
Protein only	368.8	0.124	60.8	22.94
Water only	115.8	0.212	169.0	73.91

The protein contribution reproduces most of the differential variance and the short lifetime. Water alone has a much smaller differential amplitude and a substantially longer T_2^*, consistent with slower, more common-mode fluctuations. The all-MM site-solvation first crossings are 78.5 and 71.6 fs, whereas the coherence envelope reaches e^{-1} at 21.8 fs. The classical variance diagnostic $\sigma_{A-B}^2/(2k_\text{B}T)$ is 355 cm^{-1} at 300 K; it is not interpreted as a quantum reorganization energy.

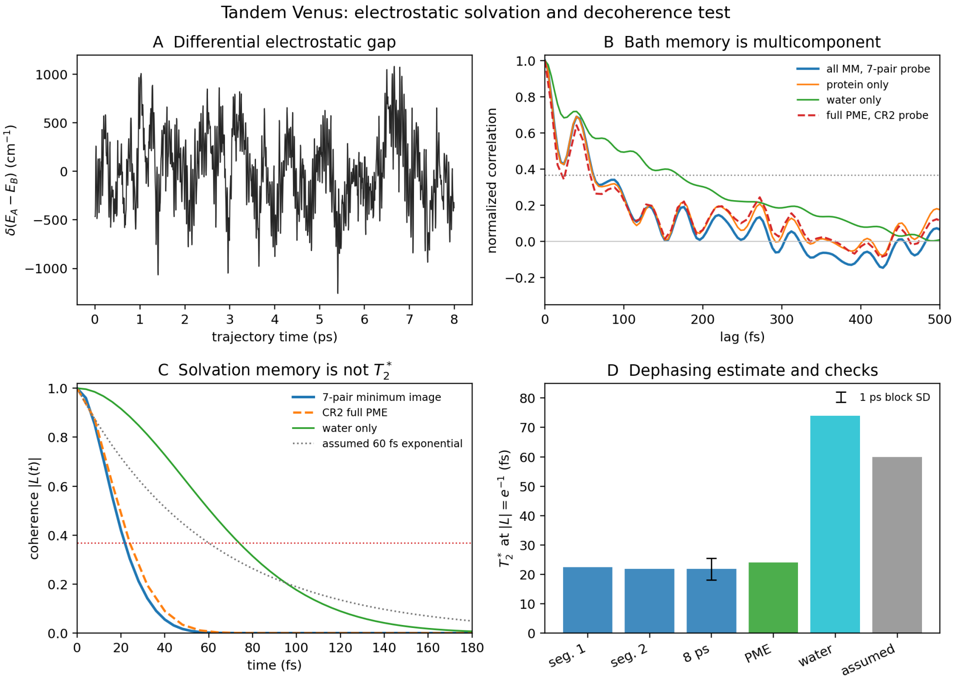

Figure 1: Project-generated validation summary. The panels show the differential gap, multicomponent correlation, cumulant coherence envelopes, and segment/block/PME/component lifetime checks.

An independent full-PME calculation perturbed the neutralized CR2 difference probe on the actual chromophore atoms and differentiated the OpenMM PME energy. At 8 fs sampling it gives $T_2^* = 24.0146$ fs. The PME and matched minimum-image differential traces correlate at 0.9736, and the corresponding minimum-image lifetime is 23.0049 fs. The PME site-memory crossings are 71.1 and 62.0 fs. Its differential correlation is oscillatory: it first crosses $1/e$ at 21.5 fs, rebounds, and has a 74.3 fs positive-integral time. A single exponential "solvation time" is therefore not an adequate description.

4.1 Robustness checks

Table 2: Sensitivity and replication checks for the classical cumulant lifetime.

Check	Result
Two contiguous 4 ps segments	22.40 and 21.81 fs
Eight independent 1 ps blocks	median 24.52 fs; SD 3.73 fs; range 19.67–29.99 fs
Sampling at 4, 8, 16 and 32 fs	21.80, 22.24, 24.21 and 22.52 fs
Remove explicit ions from estimator	21.80 to 21.62 fs
Dominant NTO pair only	21.95 fs
Dipole-matched seven-pair atomic probe	22.01 fs
Full PME cross-check	24.01 fs
PME finite-difference scale halved	0.00055 cm^{-1} RMS trace change

These checks show that the 22–24 fs result is not caused by sampling stride, one trajectory segment, the small ion contribution, NTO truncation, probe dipole mismatch or the minimum-image electrostatic kernel.

5 Spectral sensitivity with the corrected coupling

For an exponential homogeneous coherence envelope, the Lorentzian half-width is

$$\Gamma_{\mathrm{HWHM}} = \frac{1}{2\pi c T_2^*}. \tag{7}$$

The manuscript's 60 fs reference gives 88.48 cm^{-1}. Using the PME lifetime gives 221.07 cm^{-1} (FWHM 442.13 cm^{-1}), which exceeds the mean Davydov splitting $2J = 65.63$ cm^{-1} for $J =$ 32.82 cm^{-1}. The exciton centres remain at 522.98 and 524.79 nm because changing T_2^* changes broadening rather than the coupling.

The CD sign order survives, but its extrema move from 522.30/525.47 nm to 520.32/527.49 nm. More importantly, the raw peak falls to 0.173 of the 60 fs value, an 82.7% reduction. Independent normalization conceals this loss; Fig. 2 therefore uses one common amplitude scale.

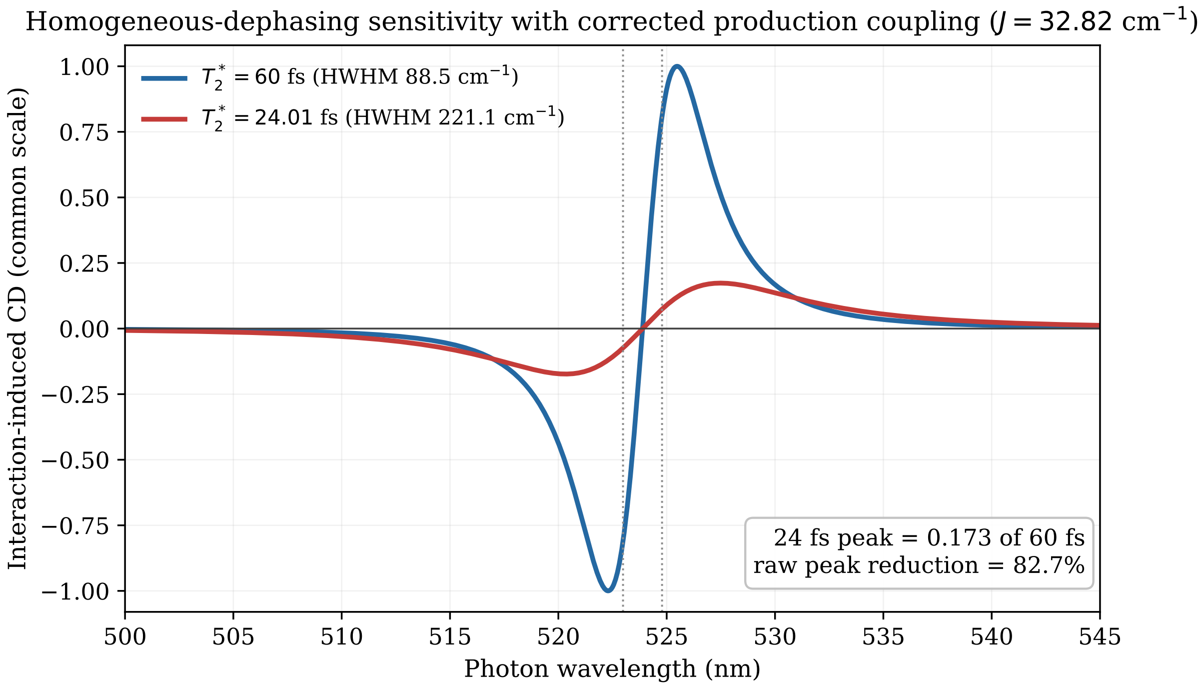

Figure 2: Project-generated 60 fs and 24.01 fs tandem CD differences on a shared amplitude scale. Dotted lines mark the mean exciton centres. The faster-dephasing curve is broader and has only 17.3% of the raw 60 fs peak.

Under the simple criterion FWHM $< 2J$, the threshold for resolving the two branches is $T_2^* = 161.7$ fs. A 22–24 fs Lorentzian lifetime would therefore not support a resolved linear-absorption doublet, even though the interaction-induced CD sign topology remains visible.

6 Relation to the experimental literature

Nguyen et al. report a constrained dVenus apparent coupling of 130.79 cm^{-1} and Lorentzian HWHM parameters of 293.51 and 439.25 cm^{-1} in their three-component difference-CD fit [2]. The corrected production value used for this sensitivity curve is $J = 32.82 \pm 1.55$ cm^{-1}, about fourfold below the constrained apparent coupling. The 24 fs model width is directionally closer to the experimental fit widths than the 60 fs width, but this is not a direct validation: the experimental widths also contain inhomogeneous, vibronic and site-energy contributions absent from the present S1-only model.

Nguyen et al. also report a 128.0 ± 8.1 fs coherence time for a dried dVenus-TD thin film measured with two-pulse near-field microscopy [2]. That experiment probes an optical ground–excited-state polarization in a thin film. The present calculation probes differential inter-site phase noise in an explicitly solvated dimer. The samples and observables differ, so their numerical values must not be equated.

Kim et al. observed a room-temperature Venus CD couplet and photon antibunching, but did not directly measure a Venus T_2^*[1]. Their 0.44–44 ps range was inferred from an assumed energy-transfer rate and coupling regime. The 22–24 fs classical estimate is much shorter than that inference. It remains compatible with the direct steady-state observation in the limited sense that vertical absorption occurs on a roughly 1–10 fs timescale and can imprint an excitonic spectrum before subsequent phase loss. The calculation therefore supports rapid optical imprinting while challenging a simple picture of the beta barrel as a passive source of long-lived coherence: the barrel can shield common-mode bulk-water fluctuations while its internal protein electrostatics remain an active differential bath.

7 Limitations and publication threshold

1. Equation (6) uses a classical fixed-charge correlation and lacks quantum detailed balance. It is not a rigorous quantum T_2^*.
2. The probe is a fixed STEOM difference density. Geometry-dependent excitation energies, intrachromophore vibrations, non-Condon effects and population relaxation are omitted.
3. AMBER/TIP3P is non-polarizable; mutual QM/MM induction is absent.
4. Eight picoseconds resolves the ultrafast decay but cannot converge slow protein and solvent modes.
5. The retained production box is net $-2e$ after transplanting two exact -1 CR2 parameter sets. Removing ions changes T_2^* by only 0.2 fs, but a publication-quality calculation should rebuild and neutralize the box.
6. The stable first crossings, block replication and direct coherence envelope are preferred over unstable biexponential tail fits.

A publication-quality estimate should compute geometry-dependent electrostatic-embedding QM/MM gaps on a high-cadence subset, calibrate the tractable excited-state method against retained STEOM points, construct a quantum-corrected spectral density, and propagate a non-Markovian lineshape or reduced-density-matrix model. Until then, 22–24 fs should be described as a classical electrostatic sensitivity result.

8 Reproducibility map

The difference probe is constructed by `build_steom_difference_probe.py`. Local electrostatic traces and the classical cumulant are generated by `analyze_solvation_decoherence.py`; the independent PME derivative is implemented in `analyze_pme_decoherence.py`; block and stride tests are performed by `validate_solvation_decoherence.py`; and the validation summary is drawn by `plot_solvation_validation.py`. The shared-scale spectrum in Fig. 2 is generated by `notes/make_decoherence_note_figure.py`. Compact gap traces, summaries and audit records accompany these programs. Full trajectories and licensed electronic-structure scratch files remain external because of their size and redistribution constraints.

References

[1] Y. Kim, H. L. Puhl, E. Chen, G. H. Taumoefolau, T. A. Nguyen, D. S. Kliger, P. S. Blank and S. S. Vogel, "Venus$_{\mathrm{A206}}$ Dimers Behave Coherently at Room Temperature," *Biophysical Journal* **116**, 1918–1930 (2019), doi:10.1016/j.bpj.2019.04.014.

[2] T. Nguyen, H. L. Puhl, E. Chen, K. Hines, C. Rani, P. S. Blank, B. Carlotti, Y. Kim, T. Goodson III and S. S. Vogel, "Anomalous photophysical behaviors attributed to excitonic coupling in fluorescent proteins," *Biophysical Journal* **124**, 4293–4309 (2025), doi:10.1016/j.bpj.2025.10.022.